JEE (Main) Official Syllabus & Empirical Chapter Weightage

ISSUED BY NATIONAL TESTING AGENCY (NTA) • APEX BOARD FOR ENGINEERING ADMISSIONS

Examination Standard: Total Marks: 300 | Total Questions: 90 (75 to Attempt) | Duration: 180 Minutes
Section Architecture: 25 Questions per Subject (20 Section A MCQs + 5 out of 10 Section B Numerical Value Questions).
Marking Scheme: +4 for Correct, -1 for Incorrect (both MCQs & NVQs), 0 for Unattempted.
Empirical Weightage Basis: Derived across 186 official computer-based shifts and verified against latest NTA Information Bulletin.

Subject: Physics (20 Official Units/Chapters)

Chapter / Unit	Class	Official Status	Qs & Marks	Priority Tier	Prescribed Authorized Topics (NTA)
Units, Dimensions & Errors Physics and Measurement	Class 11	Active	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	SI units, fundamental and derived units; system of units; dimensions of physical quantities, dimensional analysis and its applications; least count, significant figures; errors in measurement, combination of errors, percentage error.
Motion in 1D & 2D (Projectile Motion) Kinematics	Class 11	Active	**1 - 2 Qs** 4 - 8 Marks	**High Yield**	Frame of reference, motion in a straight line, position-time graphs, speed and velocity; uniform and non-uniform motion, average speed and instantaneous velocity, uniformly accelerated motion, velocity-time and position-time graphs, relations for uniformly accelerated motion; vectors, resolution of vectors, scalar and vector products, motion in a plane, projectile motion, uniform circular motion.
Newton's Laws of Motion & Friction Laws of Motion	Class 11	Active	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Force and inertia, Newton's first law of motion; momentum, Newton's second law of motion; impulse; Newton's third law of motion; law of conservation of linear momentum and its applications; equilibrium of concurrent forces; static and kinetic friction, laws of friction, rolling friction, lubrication; dynamics of uniform circular motion, centripetal force and its applications (vehicle on level circular road, vehicle on banked road).
Work, Energy, Power & Collisions Work, Energy, and Power	Class 11	Active	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Work done by a constant force and a variable force; kinetic and potential energies, work-energy theorem, power; potential energy of a spring, conservation of mechanical energy, conservative and non-conservative forces; motion in a vertical circle; elastic and inelastic collisions in one and two dimensions.
Centre of Mass & Rotational Dynamics Rotational Motion	Class 11	Active (Rolling details rationalized)	**1 - 2 Qs** 4 - 8 Marks	**High Yield**	Centre of mass of a two-particle system, centre of mass of a rigid body; basic concepts of rotational motion; moment of a force, torque, angular momentum, conservation of angular momentum and its applications; moment of inertia, radius of gyration, values of moments of inertia for simple geometrical objects, parallel and perpendicular axes theorems and their applications; rigid body rotation about fixed axis.
Gravitation & Satellites Gravitation	Class 11	Active	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	The universal law of gravitation; acceleration due to gravity and its variation with altitude and depth; Kepler's laws of planetary motion; gravitational potential energy, gravitational potential; escape velocity, orbital velocity of a satellite, time period and energy of a satellite.
Mechanical Properties of Solids & Fluids Properties of Solids and Liquids	Class 11	Active	**2 - 3 Qs** 8 - 12 Marks	**High Yield**	Elastic behaviour, stress-strain relationship, Hooke's law, Young's modulus, bulk modulus, modulus of rigidity; pressure due to a fluid column, Pascal's law and its applications; effect of gravity on fluid pressure; viscosity, Stokes' law, terminal velocity, streamline and turbulent flow, critical velocity, Bernoulli's principle and applications; surface energy and surface tension, angle of contact, excess pressure across curved surface, capillary rise; heat, temperature, thermal expansion, specific heat capacity, calorimetry, change of state, latent heat, heat transfer - conduction, convection and radiation, Newton's law of cooling.
Thermodynamics Thermodynamics	Class 11	Active (Carnot Engine deleted by NTA)	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Thermal equilibrium, zeroth law of thermodynamics, concept of temperature; heat, work and internal energy; first law of thermodynamics, isothermal and adiabatic processes; second law of thermodynamics, reversible and irreversible processes.

Chapter / Unit	Class	Official Status	Qs & Marks	Priority Tier	Prescribed Authorized Topics (NTA)
Kinetic Theory of Gases Kinetic Theory of Gases	Class 11	Active	**1 Q** 4 Marks	**High Yield**	Equation of state of a perfect gas, work done on compressing a gas; kinetic theory of gases - assumptions, concept of pressure; kinetic energy and temperature; RMS speed of gas molecules; degrees of freedom, law of equipartition of energy and applications to specific heat capacities of gases; mean free path, Avogadro's number.
Simple Harmonic Motion & Waves Oscillations and Waves	Class 11	Active (Damped oscillations & Doppler effect deleted)	**1 - 2 Qs** 4 - 8 Marks	**High Yield**	Periodic motion - period, frequency, displacement as a function of time; periodic functions; simple harmonic motion (SHM) and its equation; phase; oscillations of a spring - restoring force and force constant; energy in SHM - kinetic and potential energies; simple pendulum - derivation of expression for its time period; wave motion, longitudinal and transverse waves, speed of a travelling wave, displacement relation for a progressive wave; principle of superposition of waves, reflection of waves, standing waves in strings and organ pipes, fundamental mode and harmonics, beats.
Electric Charges, Fields & Capacitance Electrostatics	Class 12	Active (Van de Graaff generator deleted)	**2 - 3 Qs** 8 - 12 Marks	**Guaranteed Core**	Electric charges: Conservation of charge, Coulomb's law forces between two point charges, forces between multiple charges; superposition principle and continuous charge distribution; electric field, electric field lines, electric dipole, electric field due to a dipole, torque on a dipole in a uniform electric field; electric flux, Gauss's law and its applications to find field due to infinitely long uniformly charged straight wire, uniformly charged infinite plane sheet and uniformly charged thin spherical shell; electric potential and potential difference, electric potential due to a point charge, electric dipole and system of charges; equipotential surfaces, electrical potential energy of a system of two point charges and of electric dipoles in an electrostatic field; conductors and dielectrics, capacitance and capacitors, combination of capacitors in series and in parallel, capacitance of a parallel plate capacitor with and without dielectric medium between the plates, energy stored in a capacitor.
Current Electricity & Circuits Current Electricity	Class 12	Active (Potentiometer & Resistor colour codes deleted)	**2 Qs** 8 Marks	**Guaranteed Core**	Electric current, drift velocity, mobility and their relation with electric current; Ohm's law, electrical resistance, V-I characteristics, electrical resistivity and conductivity; series and parallel combinations of resistors; temperature dependence of resistance; internal resistance of a cell, potential difference and EMF of a cell, combination of cells in series and in parallel; Kirchhoff's laws and simple applications; Wheatstone bridge, Metre bridge.
Magnetism & Magnetic Properties of Matter Magnetic Effects of Current and Magnetism	Class 12	Active (Cyclotron & Earth's magnetic elements deleted)	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Biot-Savart law and its application to current carrying circular loop; Ampere's law and its applications to infinitely long current carrying straight wire and solenoid; force on a moving charge in uniform magnetic and electric fields, Lorentz force; force on a current-carrying conductor in a uniform magnetic field; force between two parallel current-carrying conductors - definition of ampere; torque experienced by a current loop in uniform magnetic field; moving coil galvanometer, its current sensitivity and conversion to ammeter and voltmeter; current loop as a magnetic dipole and its magnetic dipole moment; magnetic field lines; magnetic properties of materials - diamagnetic, paramagnetic and ferromagnetic substances with examples.
EMI & Alternating Current Electromagnetic Induction and Alternating Currents	Class 12	Active	**2 Qs** 8 Marks	**Guaranteed Core**	Electromagnetic induction: Faraday's law, induced EMF and current; Lenz's law, eddy currents; self and mutual inductance; alternating currents, peak and RMS value of alternating current/voltage; reactance and impedance; LCR series circuit, resonance, quality factor; power in AC circuits, wattless current; AC generator and transformer.
Electromagnetic Waves Electromagnetic Waves	Class 12	Active	**1 Q** 4 Marks	**Guaranteed Core**	Displacement current; electromagnetic waves and their characteristics, transverse nature of electromagnetic waves; electromagnetic spectrum (radio waves, microwaves, infrared, visible, ultraviolet, X-rays, gamma rays) including elementary facts about their uses.

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Ray Optics & Wave Optics Optics	Class 12	Active (Resolving power & Polarisation deleted)	**2 - 3 Qs** 8 - 12 Marks	**Guaranteed Core**	Ray Optics: Reflection of light, spherical mirrors, mirror formula; refraction of light, total internal reflection and its applications, optical fibres; refraction at spherical surfaces, lenses, thin lens formula, lens maker's formula, magnification, power of a lens, combination of thin lenses in contact, refraction of light through a prism; optical instruments - microscopes and astronomical telescopes (reflecting and refracting) and their magnifying powers. Wave Optics: Wavefront and Huygens' principle, reflection and refraction of plane wave at a plane surface using wavefronts; proof of laws of reflection and refraction using Huygens' principle; interference, Young's double slit experiment and expression for fringe width, coherent sources; diffraction due to a single slit, width of central maximum.
Photoelectric Effect & Matter Waves Dual Nature of Matter and Radiation	Class 12	Active (Davisson-Germer experiment deleted)	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Dual nature of radiation, photoelectric effect, Hertz and Lenard's observations; Einstein's photoelectric equation; particle nature of light; matter waves, wave nature of particle, de Broglie relation.
Atomic Models & Nuclear Physics Atoms and Nuclei	Class 12	Active (Radioactivity decay law deleted by NTA)	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Alpha-particle scattering experiment; Rutherford's model of atom; Bohr model, energy levels, hydrogen spectrum; composition and size of nucleus, atomic masses, isotopes, isobars, isotones; mass-energy relation, mass defect; binding energy per nucleon and its variation with mass number, nuclear fission and fusion.
Semiconductor Electronics Electronic Devices	Class 12	Active (Transistors, Logic Gates & Zener Diode deleted)	**1 Q** 4 Marks	**High Yield**	Energy bands in solids (conductors, insulators and semiconductors); intrinsic and extrinsic semiconductors, p-n junction, semiconductor diode: I-V characteristics in forward and reverse bias; diode as a rectifier (half-wave and full-wave).
Physics Practicals & Instrumentation Experimental Skills	Class 12	Active (Explicitly expanded by NTA)	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Familiarity with the basic approach and observations of the experiments: Vernier calipers, Screw gauge, Simple Pendulum, Meter Scale, Young's modulus by Searle's method, Surface tension by capillary rise, Coefficient of Viscosity by terminal velocity, Speed of sound by resonance tube, Specific heat capacity by calorimeter, Resistivity using metre bridge, Resistance by Ohm's law, Focal length of convex lens and concave mirror, Angle of minimum deviation for prism, Refractive index of glass slab by travelling microscope, Characteristics of p-n junction diode, Identification of diode, LED, resistor, capacitor.

Subject: Chemistry (20 Official Units/Chapters)

Chapter / Unit	Class	Official Status	Qs & Marks	Priority Tier	Prescribed Authorized Topics (NTA)
Some Basic Concepts of Chemistry (Mole Concept) Physical Chemistry	Class 11	Active	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Matter and its nature, Dalton's atomic theory: concept of atom, molecule, element, and compound. Laws of chemical combination; atomic and molecular masses, mole concept, molar mass, percentage composition, empirical and molecular formulae; chemical reactions, stoichiometry, and calculations based on stoichiometry (limiting reagent).
Structure of Atom Physical Chemistry	Class 11	Active (Thomson & Rutherford historical models deleted)	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Nature of electromagnetic radiation, photoelectric effect; spectrum of hydrogen atom. Bohr model of a hydrogen atom - its postulates, derivation of the relations for energy of the electron and radii of the different orbits, limitations of Bohr's model; dual nature of matter, de Broglie's relationship, Heisenberg uncertainty principle. Elementary ideas of quantum mechanics, quantum mechanics, the quantum mechanical model of the atom, its important features. Concept of atomic orbitals as one-electron wave functions; variation of Ψ and Ψ^2 with r for $1s$ and $2s$ orbitals; various quantum numbers (n, l, m_l, m_s) and their significance; shapes of s, p and d orbitals, electron spin and spin quantum number; rules for filling electrons in orbitals - Aufbau principle, Pauli's exclusion principle and Hund's rule, electronic configuration of elements, extra stability of half-filled and completely filled orbitals.

Chapter / Unit	Class	Official Status	Qs & Marks	Priority Tier	Prescribed Authorized Topics (NTA)
Chemical Bonding and Molecular Structure Physical Chemistry	Class 11	Active	**2 - 3 Qs** 8 - 12 Marks	**Guaranteed Core**	Kossel-Lewis approach to chemical bond formation, the concept of ionic and covalent bonds. Ionic Bonding: Formation of ionic bonds, factors affecting the formation of ionic bonds; calculation of lattice enthalpy. Covalent Bonding: Concept of electronegativity, Fajan's rule, dipole moment; Valence Shell Electron Pair Repulsion (VSEPR) theory and shapes of simple molecules. Quantum mechanical approach to covalent bonding: Valence bond theory - its important features, the concept of hybridization involving s, p and d orbitals; resonance. Molecular Orbital Theory - Its important features, LCAOs, types of molecular orbitals (bonding, antibonding), sigma and pi-bonds, molecular orbital electronic configurations of homonuclear diatomic molecules, the concept of bond order, bond length, and bond energy. Elementary idea of metallic bonding. Hydrogen bonding and its applications.
Chemical Thermodynamics Physical Chemistry	Class 11	Active	**2 Qs** 8 Marks	**Guaranteed Core**	Fundamentals of thermodynamics: System and surroundings, extensive and intensive properties, state functions, types of processes. The first law of thermodynamics - concept of work, heat, internal energy, and enthalpy, heat capacity, molar heat capacity; Hess's law of constant heat summation; Enthalpies of bond dissociation, combustion, formation, atomization, sublimation, phase transition, hydration, ionization, and solution. The second law of thermodynamics - Spontaneity of processes; ΔS of the universe and ΔG of the system as criteria for spontaneity. ΔG° (Standard Gibbs energy change) and equilibrium constant.
Solutions Physical Chemistry	Class 12	Active	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Different methods for expressing the concentration of solution - molality, molarity, mole fraction, percentage (by volume and mass both), the vapour pressure of solutions and Raoult's Law - Ideal and non-ideal solutions, vapour pressure-composition, plots for ideal and non-ideal solutions; Colligative properties of dilute solutions - a relative lowering of vapour pressure, depression of freezing point, the elevation of boiling point and osmotic pressure; Determination of molecular mass using colligative properties; Abnormal value of molar mass, van't Hoff factor and its significance.
Equilibrium (Chemical & Ionic) Physical Chemistry	Class 11	Active	**2 - 3 Qs** 8 - 12 Marks	**Guaranteed Core**	Meaning of equilibrium, the concept of dynamic equilibrium. Equilibria involving physical processes: Solid-liquid, liquid-gas and solid-gas equilibria, Henry's law. General characteristics of equilibrium involving physical processes. Equilibrium involving chemical processes: Law of chemical equilibrium, equilibrium constants (K_p and K_c) and their significance, the significance of ΔG and ΔG° in chemical equilibrium, factors affecting equilibrium concentration, pressure, temp, the effect of catalyst; Le Chatelier's principle. Ionic equilibrium: Weak and strong electrolytes, ionization of electrolytes, various concepts of acids and bases (Arrhenius, Bronsted-Lowry and Lewis) and their ionization, acid-base equilibria (including multistage ionization) and ionization constants, ionization of water, pH scale, common ion effect, hydrolysis of salts and pH of their solutions, the solubility of sparingly soluble salts and solubility products, buffer solutions.
Redox Reactions and Electrochemistry Physical Chemistry	Class 12	Active	**2 Qs** 8 Marks	**Guaranteed Core**	Electronic concepts of oxidation and reduction, redox reactions, oxidation number, rules for assigning oxidation number, balancing of redox reactions. Electrolytic and galvanic cells, different types of electrodes, electrode potentials including standard electrode potential, half-cell and cell reactions, EMF of a Galvanic cell and its measurement; Nernst equation and its applications; Relationship between cell potential and Gibbs' energy change; Dry cell and lead accumulator; Fuel cells; Corrosion and its prevention. Electrolytic conduction, factors affecting electrolytic conduction, specific, equivalent and molar conductivities and their variation with concentration, Kohlrausch's law and its applications.
Chemical Kinetics Physical Chemistry	Class 12	Active	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Rate of a chemical reaction, factors affecting the rate of reactions: concentration, temperature, pressure, and catalyst; elementary and complex reactions, order and molecularity of reactions, rate law, rate constant and its units, differential and integrated rate equations, and half-life for zero and first-order reactions, the effect of temperature on the rate of reactions, Arrhenius theory, activation energy and its calculation, collision theory of bimolecular gaseous reactions (no derivation).

Chapter / Unit	Class	Official Status	Qs & Marks	Priority Tier	Prescribed Authorized Topics (NTA)
Classification of Elements & Periodicity Inorganic Chemistry	Class 11	Active	**1 Q** 4 Marks	**Guaranteed Core**	Modem periodic law and present form of the periodic table, s, p, d and f block elements, periodic trends in properties of elements atomic and ionic radii, ionization enthalpy, electron gain enthalpy, valence, oxidation states, and chemical reactivity.
p-Block Elements (Groups 13 to 18) Inorganic Chemistry	Class 11	Active (Rationalized to periodic trends only)	**1 - 2 Qs** 4 - 8 Marks	**High Yield**	Group 13 to Group 18 Elements. General Introduction: Electronic configuration and general trends in physical and chemical properties of elements across the periods and down the groups; unique behaviour of the first element in each group.
d- and f-Block Elements Inorganic Chemistry	Class 12	Active	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Transition Elements: General introduction, electronic configuration, occurrence and characteristics, general trends in properties of the first-row transition elements - physical properties, ionization enthalpy, oxidation states, and chemical reactivity, atomic and ionic radii, colour, catalytic behaviour, magnetic properties, complex formation, interstitial compounds, alloy formation; Preparation, properties, and uses of $K_2Cr_2O_7$ and $KMnO_4$. Inner Transition Elements: Lanthanoids - Electronic configuration, oxidation states, and lanthanoid contraction. Actinoids - Electronic configuration and oxidation states.
Coordination Compounds Inorganic Chemistry	Class 12	Active	**2 - 3 Qs** 8 - 12 Marks	**Guaranteed Core**	Introduction to coordination compounds. Werner's theory; ligands, coordination number, denticity. chelation, IUPAC nomenclature of mononuclear coordination compounds, isomerism; Bonding-Valence bond approach and basic ideas of Crystal field theory, colour and magnetic properties; Importance of co-ordination compounds (in qualitative analysis, extraction of metals and in biological systems).
Purification & Characterisation of Organic Compounds Organic Chemistry	Class 11	Active	**1 Q** 4 Marks	**High Yield**	Purification - Crystallization, sublimation, distillation, differential extraction, and chromatography - principles and their applications. Qualitative analysis - Detection of nitrogen, sulfur, phosphorus, and halogens (Lassaigne's test). Quantitative analysis (basic principles only) - Estimation of carbon, hydrogen, nitrogen, halogens, sulfur, phosphorus (Dumas, Kjeldahl, Carius methods). Calculations of empirical formulae and molecular formulae: Numerical problems in organic quantitative analysis.
Some Basic Principles of Organic Chemistry (GOC) Organic Chemistry	Class 11	Active	**2 - 3 Qs** 8 - 12 Marks	**Guaranteed Core**	Tetravalency of carbon; shapes of simple molecules - hybridization (s and p); classification of organic compounds based on functional groups: and those with cyclic, and aromatic groups; nomenclature (Trivial and IUPAC); electronic displacements in a covalent bond: inductive, electromeric, resonance, and hyperconjugation. Common types of organic reactions: Substitution, addition, elimination, and rearrangement.
Hydrocarbons (Alkanes, Alkenes, Alkynes, Arenes) Organic Chemistry	Class 11	Active	**2 Qs** 8 Marks	**Guaranteed Core**	Classification, isomerism, IUPAC nomenclature, general methods of preparation, properties, and reactions. Alkanes - Conformations: Sawhorse and Newman projections (of ethane); Mechanism of halogenation of alkanes. Alkenes - Geometrical isomerism; Mechanism of electrophilic addition: addition of hydrogen, halogens, water, hydrogen halides (Markownikoff's and peroxide effect); Ozonolysis and polymerization. Alkynes - Acidic character; Addition of hydrogen, halogens, water, and hydrogen halides; Polymerization. Aromatic hydrocarbons - Nomenclature, benzene - structure and aromaticity; Mechanism of electrophilic substitution: halogenation, nitration, Friedel-Craft's alkylation and acylation, directive influence of the functional group in mono-substituted benzene.
Haloalkanes and Haloarenes Organic Chemistry	Class 12	Active	**1 - 2 Qs** 4 - 8 Marks	**High Yield**	General methods of preparation, properties, and reactions; Nature of C-X bond; Mechanisms of substitution reactions (SN1 and SN2). Uses; Environmental effects of chloroform, iodoform, freons, and DDT.

Chapter / Unit	Class	Official Status	Qs & Marks	Priority Tier	Prescribed Authorized Topics (NTA)
Oxygen Containing Compounds (Alcohols, Phenols, Ethers, Carbonyls, Carboxylic Acids) Organic Chemistry	Class 12	Active	**3 - 4 Qs** 12 - 16 Marks	**Guaranteed Core**	Alcohols: Preparation, physical and chemical properties; Identification of primary, secondary, and tertiary alcohols (Lucas test); mechanism of dehydration. Phenols: Preparation, properties, acidic character, electrophilic substitution reactions (halogenation, nitration, sulfonation, Reimer-Tiemann, Kolbe reactions). Ethers: Preparation (Williamson synthesis) and properties. Aldehydes and Ketones: Preparation, nucleophilic addition reactions, reactivity of alpha hydrogen, aldol condensation, Cannizzaro reaction, Haloform reaction. Carboxylic Acids: Acidic strength and factors affecting it, preparation and reactions.
Nitrogen Containing Organic Compounds (Amines & Diazonium Salts) Organic Chemistry	Class 12	Active	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Amines: Preparation, classification, structure, basic character, and identification of primary, secondary, and tertiary amines (Hinsberg's test). Diazonium Salts: Importance in synthetic organic chemistry (Sandmeyer reaction, Gattermann reaction, coupling reactions).
Biomolecules Organic Chemistry	Class 12	Active	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Carbohydrates - Classification, aldoses and ketoses; monosaccharides (glucose and fructose) and constituent of oligosaccharides (sucrose, lactose, maltose). Proteins - Elementary idea of $■lpha$-amino acids, peptide bond, polypeptide; Proteins: primary, secondary, tertiary, and quaternary structure (qualitative idea only), denaturation of proteins, enzymes. Vitamins - Classification and functions. Nucleic Acids - Chemical constitution of DNA and RNA. Biological functions of nucleic acids.
Principles Related to Practical Chemistry Practical Chemistry	Class 12	Active (High Yield Section B)	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Detection of extra elements (Nitrogen, Sulfur, halogens) in organic compounds; Detection of functional groups: hydroxyl, carbonyl, carboxyl, amino. Chemistry involved in titrations: Mohr's salt, oxalic acid vs KMnO4. Qualitative analysis of cations (Pb2+, Cu2+, Fe3+, Al3+, Zn2+, Ni2+, Ca2+, Ba2+, Mg2+, NH4+) and anions (CO3 2-, S 2-, SO3 2-, SO4 2-, NO2 -, NO3 -, Cl-, Br-, I-). Chemical principles in preparation of Mohr's salt, potash alum, acetanilide.

Subject: Mathematics (14 Official Units/Chapters)

Chapter / Unit	Class	Official Status	Qs & Marks	Priority Tier	Prescribed Authorized Topics (NTA)
Sets, Relations & Functions Sets, Relations and Functions	Class 11	Active	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Sets and their representation: Union, intersection, and complement of sets and their algebraic properties; Power set; Relation, type of relations, equivalence relations; functions; one-one, into and onto functions, the composition of functions.
Complex Numbers & Quadratic Equations Complex Numbers and Quadratic Equations	Class 11	Active (Square root & polar form simplified)	**2 Qs** 8 Marks	**Guaranteed Core**	Complex numbers as ordered pairs of reals, Representation of complex numbers in the form $a + ib$ and their representation in a plane, Argand diagram, algebra of complex number, modulus and argument of a complex number; Quadratic equations in real and complex number system and their solutions; Relation between roots and coefficients, nature of roots, the formation of quadratic equations with given roots.
Matrices and Determinants Matrices and Determinants	Class 12	Active (Properties of determinants rationalized)	**2 Qs** 8 Marks	**Guaranteed Core**	Matrices, algebra of matrices, type of matrices, determinants, and matrices of order two and three, evaluation of determinants, area of triangles using determinants, Adjoint, and evaluation of inverse of a square matrix using adjoint and determinants, test of consistency and solution of simultaneous linear equations in two or three variables using matrix inversion / Cramer's rule.
Permutations & Combinations Permutations and Combinations	Class 11	Active	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	The fundamental principle of counting, permutation as an arrangement and combination as selection, Meaning of $P(n, r)$ and $C(n, r)$, simple applications.
Binomial Theorem Binomial Theorem	Class 11	Active (Properties of binomial coefficients deleted by NTA)	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Binomial theorem for a positive integral index, general term and middle term, and simple applications.

Chapter / Unit	Class	Official Status	Qs & Marks	Priority Tier	Prescribed Authorized Topics (NTA)
Sequences & Progressions (AP, GP) Sequence and Series	Class 11	Active (AGP & Special sums of squares/cubes deleted)	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Arithmetic and Geometric progressions, insertion of arithmetic, geometric means between two numbers, relation between A.M and G.M.
Differential Calculus (Limits, Continuity, AOD) Limit, Continuity and Differentiability	Class 12	Active (Rolle's & LMVT & Tangents/Normals deleted)	**3 - 4 Qs** 12 - 16 Marks	**Guaranteed Core**	Real-valued functions, algebra of functions, polynomials, rational, trigonometric, logarithmic, and exponential functions, inverse function. Graphs of simple functions. Limits, continuity, and differentiability. Differentiation of the sum, difference, product, and quotient of two functions. Differentiation of trigonometric, inverse trigonometric, logarithmic, exponential, composite and implicit functions; derivatives of order up to two. Applications of derivatives: Rate of change of quantities, monotonic - Increasing and decreasing functions, Maxima and minima of functions of one variable.
Definite & Indefinite Integrals, Area Under Curves Integral Calculus	Class 12	Active (Limit of a sum deleted)	**3 - 4 Qs** 12 - 16 Marks	**Guaranteed Core**	Integral as an anti-derivative, Fundamental integrals involving algebraic, trigonometric, exponential, and logarithmic functions. Integrations by substitution, by parts, and by partial functions. Integration using trigonometric identities. Evaluation of simple integrals. The fundamental theorem of calculus, properties of definite integrals. Evaluation of definite integrals, determining areas of the regions bounded by simple curves in standard form.
Differential Equations Differential Equations	Class 12	Active (Formation of differential equations deleted)	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Ordinary differential equations, their order, and degree. Solution of differential equations by the method of separation of variables, solution of a homogeneous and linear differential equation of the type $ rac{dy}{dx} + P(x)y = Q(x)$.
Straight Lines, Circles & Conic Sections Coordinate Geometry	Class 11	Active (Translation of axes deleted)	**3 - 4 Qs** 12 - 16 Marks	**Guaranteed Core**	Cartesian system of rectangular coordinates in a plane, distance formula, sections formula, locus, and its equation, slope of a line, parallel and perpendicular lines, intercepts of a line on the coordinate axis. Straight line: Various forms of equations of a line, intersection of lines, angles between two lines, conditions for concurrence of three lines, the distance of a point from a line, coordinate of the centroid, orthocentre, and circumcentre of a triangle. Circle, conic sections: Standard form of equations of a circle, the general form of the equation of a circle, its radius and central, equation of a circle when the endpoints of a diameter are given, points of intersection of a line and a circle with the centre at the origin and condition for a line to be tangent to a circle. Sections of conics, equations of conic sections (parabola, ellipse, and hyperbola) in standard forms, condition for $y = mx + c$ to be a tangent and point(s) of tangency.
3D Geometry (Lines Only) Three Dimensional Geometry	Class 12	Active (Planes completely deleted by NTA)	**2 Qs** 8 Marks	**Guaranteed Core**	Coordinates of a point in space, the distance between two points, section formula, directions ratios, and direction cosines, the angle between two intersecting lines. Skew lines, the shortest distance between them, and its equation. Equations of a line.
Vector Algebra Vector Algebra	Class 12	Active (Scalar & Vector Triple Products deleted)	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Vectors and scalars, the addition of vectors, components of a vector in two and three-dimensional space, scalar and vector products.
Statistics & Probability Statistics and Probability	Class 11	Active (Bernoulli trials & binomial distribution deleted)	**2 Qs** 8 Marks	**Guaranteed Core**	Measures of dispersion; calculation of mean, median, mode of grouped and ungrouped data, calculation of standard deviation, variance and mean deviation for grouped and ungrouped data. Probability: Probability of an event, addition and multiplication theorems of probability, Baye's theorem, probability distribution of a random variable.
Trigonometric Functions & Inverse Trig Trigonometry	Class 11	Active (Heights and distances deleted)	**1 - 2 Qs** 4 - 8 Marks	**Guaranteed Core**	Trigonometrical identities and equations. Trigonometrical functions. Inverse trigonometrical functions and their properties.